

4.1 Electrical System Completion Log

Completion and testing of the electrical system from rough-in through final inspection — every outlet, switch, fixture and circuit verified before the inspector arrives.

Project Address: _____

Project Name: _____ Permit Number: _____

Electrician / Company: _____ License Number: _____

Contact Phone: _____

PURPOSE: This log documents the completion and testing of all electrical systems from rough-in through final inspection. Use it to verify every outlet, switch, fixture and circuit is properly installed and functioning before final inspection.

ROUGH-IN TO FINISH VERIFICATION

Review rough-in work and verify proper completion:

- ☐ All electrical boxes properly located per plan
- ☐ All boxes properly secured to framing
- ☐ Box depth appropriate for wall finish thickness
- ☐ All boxes filled out flush with finished wall surface
- ☐ Vapor barrier properly sealed around boxes (if applicable)
- ☐ All required backing/blocking installed for fixtures
- ☐ All cables properly secured within 8" of boxes
- ☐ Cable stapling meets code (every 4.5 feet, 12" from boxes)
- ☐ Proper cable protection through studs (1-1/4" from edge or nail plates)
- ☐ No damaged cable jackets or conductors
- ☐ All wire splices made in accessible boxes only
- ☐ All boxes have required cubic inch capacity for number of conductors

DEVICE INSTALLATION VERIFICATION

- ☐
- All receptacles installed with correct orientation (ground down or up per preference)

☐☐☐☐☐☐☐☐☐☐☐☐☐☐

OUTLET TESTING LOG

Test and document each outlet/receptacle:

Receptacle Test Record

Room / Location	Outlet ID	Polarity OK	GFCI Test	Notes
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Receptacle Test Record

Room / Location	Outlet ID	Polarity OK	GFCI Test	Notes
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Receptacle Test Record

Room / Location	Outlet ID	Polarity OK	GFCI Test	Notes
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		☐	☐	

Testing notes: Use an outlet tester for polarity. Press TEST on every GFCI outlet — it should trip. Press RESET to restore power.

LIGHTING FIXTURES INSTALLATION

Fixture Installation & Test Record

Room / Location	Fixture Type	Installed	Bulb Type	Dimmer Compatible	Tested OK	Notes
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	

Fixture Installation & Test Record

Room / Location	Fixture Type	Installed	Bulb Type	Dimmer Compatible	Tested OK	Notes
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
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		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	
		☐		☐ Y ☐ N	☐	

CEILING FAN INSTALLATION & TESTING

Ceiling Fan Record

Room / Location	Fan Model	Box Rated for Fan	Light Kit	Remote / Wall Control	All Speeds Work	No Wobble	Notes
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Fan Installation Checks

- ☐ Ceiling box rated for fan weight (must be fan-rated box)
- ☐ Fan properly secured to fan-rated box or brace
- ☐ Downrod length appropriate for ceiling height
- ☐ All fan blades balanced and secure
- ☐ Light kit properly installed (if applicable)
- ☐ Wall control or remote functioning properly
- ☐ All fan speeds operate smoothly
- ☐ No wobble or unusual noise at any speed
- ☐ Reverse function works (summer/winter direction)

SWITCH OPERATION TESTING

Test every switch and verify correct fixture control:

Switch Test Record

Room / Location	Switch Location	Controls Which Fixture	Type	Works OK	Notes
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				☐	

Switch Test Record

Room / Location	Switch Location	Controls Which Fixture	Type	Works OK	Notes
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Type codes: S = single-pole · 3 = 3-way · 4 = 4-way · D = dimmer · P = pilot light · T = timer

ELECTRICAL PANEL COMPLETION

Panel Make/Model: _____ Main Breaker Size: _____

Panel Location: _____ Number of Circuits: _____

Panel Completion Checklist

- ☐ All circuit breakers properly installed and seated
- ☐ All breaker connections tight (proper torque per manufacturer)
- ☐ Main breaker correct size for service (typically 100A, 150A or 200A)
- ☐ All circuits properly labeled on breaker
- ☐ Panel directory card completely filled out and legible
- ☐ All GFCI breakers installed and functioning (test button works)
- ☐ All AFCI breakers installed per code requirements
- ☐ Combination AFCI/GFCI breakers where required
- ☐ No double-tapped breakers (unless breaker is rated for 2 wires)
- ☐ All unused breaker spaces filled with blank plates
- ☐ All knockouts sealed (no open holes in panel)
- ☐ Panel cover/dead front properly installed
- ☐ Panel cover screws all installed and tight
- ☐ Proper working clearance maintained (30" wide x 36" deep x 6'6" high)
- ☐ Panel properly grounded (ground wire to ground bar)
- ☐ Neutral and ground bars properly separated (if required)
- ☐ Service entrance cable properly secured
- ☐ Panel labeled "MAIN ELECTRICAL PANEL" or "MAIN DISCONNECT"

CIRCUIT DIRECTORY

Document all circuits for the panel directory:

Panel Directory

Breaker #	Amperage	Type	Circuit Description / Serves	GFCI	AFCI
1				☐	☐
2				☐	☐
3				☐	☐
4				☐	☐
5				☐	☐
6				☐	☐
7				☐	☐
8				☐	☐
9				☐	☐
10				☐	☐
11				☐	☐
12				☐	☐
13				☐	☐
14				☐	☐
15				☐	☐
16				☐	☐
17				☐	☐
18				☐	☐
19				☐	☐

Panel Directory

Breaker #	Amperage	Type	Circuit Description / Serves	GFCI	AFCI
20				☐	☐
21				☐	☐
22				☐	☐
23				☐	☐
24				☐	☐
25				☐	☐
26				☐	☐
27				☐	☐
28				☐	☐
29				☐	☐
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31				☐	☐
32				☐	☐
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35				☐	☐
36				☐	☐
37				☐	☐
38				☐	☐
39				☐	☐

Panel Directory

Breaker #	Amperage	Type	Circuit Description / Serves	GFCI	AFCI
40				☐	☐
41				☐	☐
42				☐	☐

Type codes: SP = single-pole · DP = double-pole (240V)

SMOKE & CO DETECTOR INSTALLATION

Detector Record

Location	Type	Make / Model	Inter-connected	Battery Backup	Tested OK	Install Date
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			☐	☐	☐	

Detector Requirements & Testing

- ☐ Smoke detector in every bedroom
- ☐ Smoke detector outside each sleeping area
- ☐ Smoke detector on every level including basement
- ☐ All smoke detectors hard-wired with battery backup
- ☐ All smoke detectors interconnected (test one, all sound)
- ☐ Carbon monoxide detector on every level
- ☐ Carbon monoxide detector near sleeping areas
- ☐ CO detectors hard-wired with battery backup (or plug-in with battery)
- ☐ All detectors tested with test button — alarm sounds
- ☐ Interconnection tested — trigger one, all alarm
- ☐ Date of installation written on detector
- ☐ Manufacturer instructions left with homeowner

SPECIAL SYSTEMS & EQUIPMENT

Kitchen Appliances

- ☐ Range/cooktop circuit — proper voltage (240V or 120V)
- ☐ Range receptacle matches appliance plug, or direct wire connection made
- ☐ Dishwasher circuit — dedicated 20A circuit
- ☐ Dishwasher receptacle or direct wire as required
- ☐ Disposal circuit — GFCI protected or on dedicated circuit
- ☐ Microwave circuit — dedicated 20A circuit (if built-in)
- ☐ Refrigerator circuit — dedicated 20A circuit
- ☐ All appliances tested and functioning

HVAC Equipment

- ☐ Furnace/air handler circuit — correct amperage
- ☐ Disconnect switch installed within sight of equipment
- ☐ Condensing unit circuit — correct amperage (typically 30–60A)
- ☐ Condensing unit disconnect installed and accessible
- ☐ Thermostat wiring complete and functioning
- ☐ All HVAC equipment properly grounded

Water Heater

- ☐ Water heater circuit — correct amperage (check data plate)
- ☐ Water heater properly wired (240V for electric)
- ☐ Disconnect or breaker lockout as required by code
- ☐ Water heater properly grounded

Laundry

- ☐ Washer circuit — dedicated 20A circuit
- ☐ Dryer circuit — 240V 30A circuit (electric dryer)
- ☐ Dryer receptacle — NEMA 10-30R or 14-30R
- ☐ All laundry area outlets GFCI protected

Garage & Exterior

- ☐ All garage receptacles GFCI protected
- ☐ Garage door opener circuit and receptacle
- ☐ Garage door opener functioning properly
- ☐ All exterior receptacles GFCI protected
- ☐ All exterior receptacles have weatherproof covers
- ☐ All exterior lighting installed and functioning
- ☐ Exterior lighting on photocell or timer (if specified)
- ☐ Landscape lighting transformer installed (if applicable)
- ☐ Service entrance lighting installed and functioning

Other Systems

System	Status	Notes
Doorbell / chime — working	☐ Yes ☐ No	
Security system — installed	☐ Yes ☐ No ☐ N/A	
Generator transfer switch	☐ Yes ☐ No ☐ N/A	
Sump pump circuit	☐ Yes ☐ No ☐ N/A	
Well pump circuit	☐ Yes ☐ No ☐ N/A	
Septic pump circuit	☐ Yes ☐ No ☐ N/A	
Attic / crawl space lights	☐ Yes ☐ No ☐ N/A	

FINAL ELECTRICAL TESTING

Complete this testing before requesting the final inspection:

- ☐ ALL outlets tested with circuit tester — correct polarity, no open grounds
- ☐ ALL GFCI outlets tested — trip function works, reset function works
- ☐ ALL switches tested — operate correct fixtures
- ☐ ALL 3-way and 4-way switches tested from all locations
- ☐ ALL light fixtures tested — all bulbs working
- ☐ ALL dimmer switches tested — smooth dimming operation
- ☐ ALL ceiling fans tested — all speeds, reverse function, light kits
- ☐ ALL smoke detectors tested — alarm sounds
- ☐ ALL smoke detectors interconnection tested — all alarm together
- ☐ ALL carbon monoxide detectors tested — alarm sounds
- ☐ ALL appliance circuits tested with appliances operating
- ☐ ALL exterior outlets and lights tested
- ☐ Garage door opener tested
- ☐ Doorbell tested

- ☐ Panel directory verified — all circuits correctly labeled
- ☐ Main disconnect operation tested
- ☐ No flickering lights when major loads turn on
- ☐ No warm or hot outlets, switches or breakers under load
- ☐ No buzzing or humming from outlets, switches or panel

FINAL ELECTRICAL INSPECTION

Inspection Requested Date: _____ Inspection Scheduled Date: _____

Inspector Name: _____

Inspection Date: _____ Inspection Time: _____

INSPECTION RESULT: ☐ PASSED — Certificate Issued ☐ FAILED — Corrections Required

Required Corrections

Corrections Completed Date: _____ Re-Inspection Date: _____

RE-INSPECTION RESULT: ☐ PASSED ☐ FAILED

Certificate of Occupancy / Electrical Certificate

Certificate Number: _____ Date Issued: _____

Issued By: _____

Sign-Off

Electrician Signature: _____ Date: _____

Owner/Builder Signature: _____ Date: _____